

JEE MAIN WEEKEND TEST SERIES



NARAYANA
IIT ACADEMY
INDIA



Sec: INC.JR.IIT _*CO-SC
Time: 3 Hrs

Date: 14-04-24
Max. Marks: 186

14-04-24_INC.JR.*CO-SC_JEE-ADV_WAT-1_SYLLABUS

- MATHEMATICS:** **PRESENTWEEK (100%): BASIC MATHEMATICS :** Number System, Definition of Intervals, Divisibility test, Ratio and proportion, Law of indices, Algebraic Identities, Inequalities Involving Linear terms, Factorisation, Wavy Curve Method, Modulus functions and their properties, Equations/Inequations of modulus functions involving only linear terms
- PHYSICS:** **PRESENTWEEK (100%): GEOMETRICAL OPTICS:** Complete Reflection of Light (Plane mirrors + Spherical mirrors)
(**Exclude:** Motion of Object & Image, Calculus based approach models)
- CHEMISTRY:** **PRESENTWEEK (100%): STOICHIOMETRY-I:** Mole , significant figures, laws of chemical combination, Atomic mass unit, molar mass, avg molar mass, Molar volume, Vapour density, Empirical & Molecular formula, Stoichiometric calculations, w/W, w/V, v/V, Percentage yield, Percentage purity, Successive reactions, Limiting reagent

Time: 3HRS

IMPORTANT INSTRUCTIONS

Max Marks: 186

PHYSICS:

Section	Question Type	+Ve Marks	- Ve Marks	No.of Qs	Total marks
Sec – I(Q.N : 1 – 5)	Questions with Single Correct Choice	3	-1	5	15
Sec – II(Q.N : 6 – 13)	Questions with Multiple Correct Choice (Partial Marking +1)	4	-2	8	32
Sec – III(Q.N : 14 – 18)	Questions with Integer Answer Type	3	0	5	15
Total				18	62

CHEMISTRY:

Section	Question Type	+Ve Marks	- Ve Marks	No.of Qs	Total marks
Sec – I(Q.N : 19 – 23)	Questions with Single Correct Choice	3	-1	5	15
Sec – II(Q.N : 24 – 31)	Questions with Multiple Correct Choice (Partial Marking +1)	4	-2	8	32
Sec – III(Q.N : 32 – 36)	Questions with Integer Answer Type	3	0	5	15
Total				18	62

MATHEMATICS:

Section	Question Type	+Ve Marks	- Ve Marks	No.of Qs	Total marks
Sec – I(Q.N : 37 – 41)	Questions with Single Correct Choice	3	-1	5	15
Sec – II(Q.N : 42 – 49)	Questions with Multiple Correct Choice (Partial Marking +1)	4	-2	8	32
Sec – III(Q.N : 50 – 54)	Questions with Integer Answer Type	3	0	5	15
Total				18	62

THE NARAYANA GROUP

MATHEMATICS

Max Marks: 62

SECTION – I

(SINGLE CORRECT ANSWER TYPE)

This section contains 5 multiple choice questions. Each question has 4 options (A), (B), (C) and (D) for its answer, out of which ONLY ONE option can be correct.

Marking scheme: +3 for correct answer, 0 if not attempted and –1 in all other cases.

1. The solution set for $\left| \frac{2(3-x)}{5} \right| < \frac{3}{5}$
- A) $\left(\frac{1}{4}, \frac{3}{4} \right)$ B) $\left(\frac{3}{2}, \frac{9}{2} \right)$ C) $\left(\frac{1}{2}, \frac{3}{2} \right)$ D) $\left(\frac{3}{2}, \frac{5}{2} \right)$
2. If x is real and $x^2 - 3x + 2 > 0$, $x^2 - 3x - 4 \leq 0$, then which of the following is correct?
- A) $-1 \leq x < 4$ B) $2 \leq x \leq 4$
C) $-1 < x \leq 1$ D) $-1 \leq x < 1$ or $2 < x \leq 4$
3. The least integer satisfying,
 $49.4 - \frac{(27-x)}{10} < 47.4 - \left(\frac{27-9x}{10} \right)$ is
- A) 2 B) 3 C) 4 D) none of these
4. The solution of the inequation $\frac{7}{(x-2)(x-3)} + \frac{9}{(x-3)} + 1 < 0$ is
- A) $(-5, 3)$ B) $(-5, 2)$ C) $(1, 2) \cup (3, \infty)$ D) $(-5, 1) \cup (2, 3)$
5. Find the number 'n' that has the first ten natural numbers as its factors and is the smallest 5 digit number
- A) 12,520 B) 10,080 C) 36,280 D) 10,040

SECTION – II

(MULTIPLE CORRECT ANSWER TYPE)

This section contains 8 multiple choice questions. Each question has 4 options (A), (B), (C) and (D) for its answer, out of which ONE OR MORE than ONE option can be correct.

Marking scheme: +4 for all correct options & +1 partial marks, 0 if not attempted and -2 in all wrong cases

6. If $\sqrt{a-x} + \sqrt{b-x} + \sqrt{c-x} = 0$, then the value of $(a+b+c+3x)(a+b+c-x)$ is
- A) $4(ab+bc+ca)$ B) $12ab$
C) $6(ab+bc)$ D) 0

7. If $\frac{|x+3|+x}{x+2} > 1$ then solution of x contains the following intervals
- A) $(-5, -2)$ B) $(-1, \infty)$ C) $(-2, -1)$ D) $(-8, -5)$
8. Which of the following statements are true
- A) Solution set of the inequality $-15 < \frac{3(x-2)}{5} \leq 0$ is $(-23, 2]$
- B) Solution set of the inequality $7 \leq \frac{3x+11}{2} \leq 11$ is $\left[1, \frac{11}{3}\right]$
- C) Solution set of the inequality $-5 \leq \frac{2-3x}{4} \leq 9$ is $[-1, 1] \cup [3, 5]$
- D) None of these
9. The solution set of $\left|\frac{x+1}{x}\right| + |x+1| = \frac{(x+1)^2}{|x|}$ contains
- A) $\{x \in \mathbb{R} / x > 0\}$ B) $\{-1\}$
- C) $\{-1, 1\}$ D) $\{x \in \mathbb{R} / x \geq 0\}$
10. A manufacturer has 600 liters of a 12% solution of acid. How many litres of a 30% acid solution must be added to it so that acid content in the resulting mixture will be more than 15% but less than 18%?
- A) More than 120 liters but less than 300 litres
- B) More than 140 liters but less than 600 litres
- C) More than 100 liters but less than 280 litres
- D) More than 160 liters but less than 500 litres

11. If a is a prime number less than 100 such that $b = a + 2$, in how many cases are both a and b primes?
A) 10 B) 8 C) 9 D) 7
12. Which of the following is a root of the equation $|2 - |1 - |x|| = 1$ is/are
A) 0 B) 2 C) -2 D) 4
13. 100 people standing in a circle in an order 1 to 100. No. 1 has a sword. He kills the next person (i.e. No. 2) and gives the sword to the next (i.e. No. 3). All people do the same until only 1 survives. Which number survives at the last? There are 100 people starting from 1 to 100 .
A) 72 B) 73 C) 74 D) 75

SECTION – III
(INTEGER ANSWER TYPE)

This section contains 5 questions. The answer is a single digit integer ranging from 0 to 9 (both inclusive).

Marking scheme +3 for correct answer, 0 if not attempted and 0 in all other cases.

14. If $\left(\sqrt[5]{4}\right)^{4x+7} = \left(\sqrt[11]{64}\right)^{2x+7}$, then the value of x is
15. If $x = \sqrt[3]{4 + \sqrt{15}} + \sqrt[3]{4 - \sqrt{15}}$, then the value of $x^3 - 3x$ is
16. Solution of $|x - 1| + |x - 2| + |x - 3| \geq 6$ is $(-\infty, K_1] \cup [K_2, \infty)$ then $K_1 + K_2$ is equal to
17. The set of real values of x satisfying $||x - 1| - 1| \leq 1$ is $[a, b]$, then $\left|\frac{b}{a}\right|$ is equal to
18. The number of integral solutions of $\frac{x+2}{x^2+1} > \frac{1}{2}$ is

SECTION – I
(SINGLE CORRECT ANSWER TYPE)

This section contains 5 multiple choice questions. Each question has 4 options (A), (B), (C) and (D) for its answer, out of which ONLY ONE option can be correct.

Marking scheme: +3 for correct answer, 0 if not attempted and –1 in all other cases.

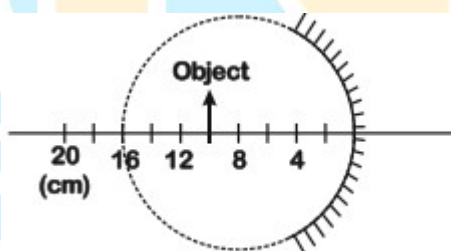
19. Two plane mirrors are inclined to each other such that a ray of light incident on the first mirror (M_1) and parallel to the second mirror (M_2) is finally reflected from the second mirror (M_2) parallel to the first mirror (M_1). The angle between the two mirrors will be

A) 75° B) 45° C) 90° D) 60°

20. A concave mirror for face viewing has focal length of 0.4 m. The distance at which you hold the mirror from your face in order to see your image upright with a magnification of 5 is

A) 0.32 m B) 0.24 m C) 1.60 m D) 0.16 m

21. A spherical mirror is obtained as shown in the figure from a hollow glass sphere. If an object is positioned in front of the mirror, what will be the nature and magnification of the image of the object? (Figure drawn as schematic and not to scale)



- A) Inverted, real and unmagnified
B) Inverted, real and magnified
C) Erect, virtual and magnified
D) Erect, virtual and unmagnified
22. The focal length f is related to the radius of curvature r of the spherical convex mirror by
- A) $f = r$ B) $f = -\frac{1}{2}r$ C) $f = +\frac{1}{2}r$ D) $f = -r$

23. An object is placed beyond the centre of curvature C of the given concave mirror. If the distance of the object is d_1 from C and the distance of the image formed is d_2 from C , the radius of curvature of this mirror is:

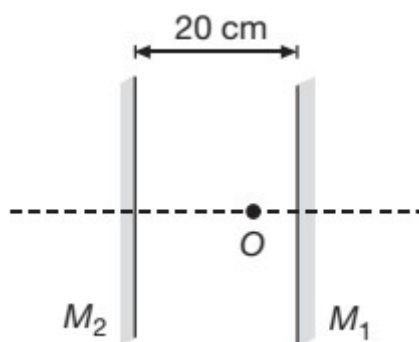
A) $\frac{d_1 d_2}{d_1 - d_2}$ B) $\frac{d_1 d_2}{d_1 + d_2}$ C) $\frac{2d_1 d_2}{d_1 + d_2}$ D) $\frac{2d_1 d_2}{d_1 - d_2}$

SECTION – II
(MULTIPLE CORRECT ANSWER TYPE)

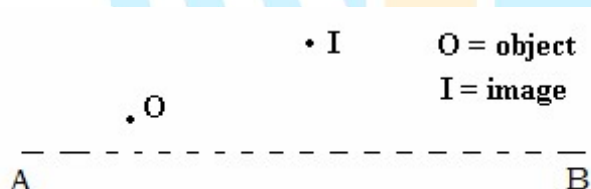
This section contains 8 multiple choice questions. Each question has 4 options (A), (B), (C) and (D) for its answer, out of which ONE OR MORE than ONE option can be correct.

Marking scheme: +4 for all correct options & +1 partial marks, 0 if not attempted and -2 in all wrong cases

24. Two plane mirrors at an angle such that a ray incident on a mirror undergoes a total deviation of 240° after two reflections.
- A) the angle between the mirror is 60°
B) the number of images formed by this system will be 5, if an object is placed symmetrically between the mirrors.
C) the no. of images will be 5 if an object is kept unsymmetrically between the mirrors.
D) a ray will retrace its path after 2 successive reflections, if the angle of incidence on one mirror is 60° .
25. A concave mirror cannot form
- A) virtual image of virtual object B) virtual image of a real object
C) real image of a real object D) real image of a virtual object.
26. Two plane mirrors M_1 and M_2 are placed parallel to each other 20cm apart. A luminous point object 'O' is placed between them at 5cm from M_1 as shown.

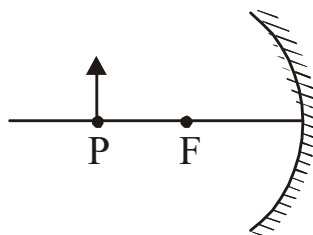


- A) The distances (in cm) of three nearest images from mirror M_1 are 5, 35 and 45 respectively.
- B) The distances (in cm) of three nearest images from mirror M_2 are 5, 35 and 45 respectively.
- C) The distances (in cm) of three nearest images from mirror M_1 are 15, 25 and 55 respectively.
- D) The distances (in cm) of three nearest images from mirror M_2 are 15, 25 and 55 respectively.
27. For a mirror, the linear magnification is +2. The conclusion(s) that can be drawn from this information is/are
- A) The mirror may be concave
- B) The mirror can be convex or concave but it cannot be plane
- C) The object lies between pole and focus
- D) The object lies beyond focus
28. In the given diagram AB is the principal axis of mirror. Mark the correct options.

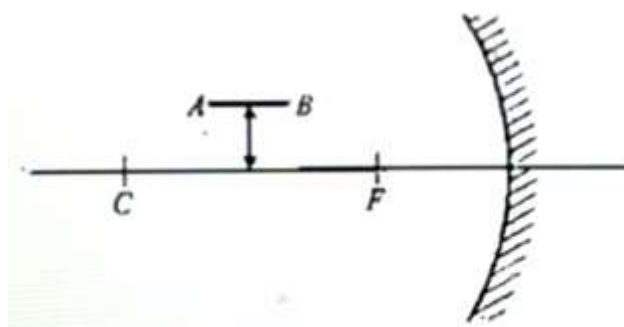


- A) If O is virtual then optical device may be a convex mirror.
- B) If O is virtual then optical device may be a concave mirror.
- C) If I is virtual then optical device may be a convex mirror.
- D) If I is virtual then optical device may be a concave mirror.

29. The diagram below shows an object located at point P, 0.25 meter from a concave spherical mirror with principal focus F. The focal length of the mirror is 0.10 meter. How does the image change as the object is moved from point P toward point F ?



- A) Its distance from the mirror decreases
B) The size of image decreases
C) Its distance from the mirror increases
D) The size of images increases
30. A two eyed man is looking at the junction of two mutually perpendicular mirrors from a far off distance. Assume no reflection to occur from the edge. Then if both the eyes are open
- A) The eye 1 of man can see image of both eye 1 and eye 2
B) The eye 1 can see image of eye 1 only and eye 2 see image of eye 2 only
C) The eye 1 can see image of eye 2 only and eye 2 can see image of eye 1 only.
D) All the above statements are false.
13. An object AB is placed parallel and close to the optical axis between focus 'F' and centre of curvature 'C' of a converging mirror of focal length ' f ' as shown in figure



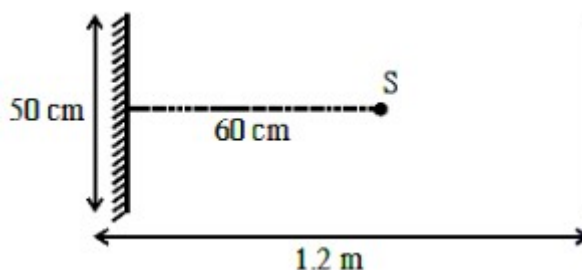
- A) Image of A will be closer than that of B from the mirror
- B) Image of AB will be parallel to the optical axis
- C) Image of AB will be straight line inclined to the optical axis
- D) Image of AB will be not straight line

SECTION – III
(INTEGER ANSWER TYPE)

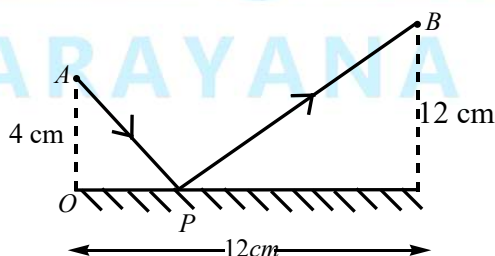
This section contains 5 questions. The answer is a single digit integer ranging from 0 to 9 (both inclusive).

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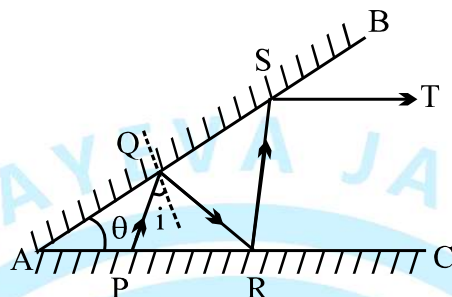
32. A point source of light S, placed at a distance 60 cm in front of the centre of a plane mirror of width 50 cm, hangs vertically on a wall. A man walks in front of the mirror along a line parallel to the mirror at a distance 1.2 m from it (see in the figure). The distance between the extreme points where he can see the image of the light source in the mirror is 25p cm. Find p.



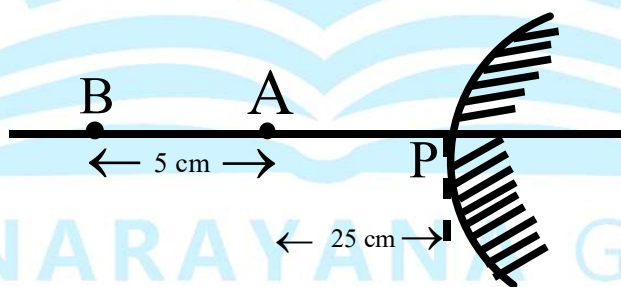
33. A ray of light is to travel from point A to point B as shown in figure after reflecting from P. Find OP (in cm).



34. Two plane mirror AB and AC are inclined at an angle of 20° . A ray of light starting from point P is incident at point Q on the mirror AB, then at R on mirror AC and again at S on AB, finally the ray ST goes parallel to the mirror AC. Let the angle be i (in degrees) which the ray makes with the normal at point Q on mirror AB. Find the value of $\frac{180}{i}$



35. A ray of light is incident at 50° on the middle of one of the two mirrors arranged at angle of 60° between them. The ray then touches the second mirror, get reflected back to the first mirror, making an angle of incidence of $10N^\circ$. Find the value of N
36. A convex mirror of focal length 10 cm is shown in the figure. A linear object AB = 5 cm is placed along the optic axis. Point A is at a distance of 25 cm from the pole of mirror. The length of image of AB is $\frac{n}{14}$ cm. Find the value of n.



SECTION – I
(SINGLE CORRECT ANSWER TYPE)

This section contains 5 multiple choice questions. Each question has 4 options (A), (B), (C) and (D) for its answer, out of which ONLY ONE option can be correct.

Marking scheme: +3 for correct answer, 0 if not attempted and –1 in all other cases.

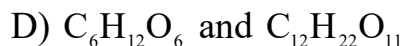
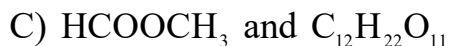
37. Mass of sucrose ($C_{12}H_{22}O_{11}$) produced by mixing 84gm of carbon, 12gm of hydrogen and 56 lit. O_2 at 1 atm & 273K according to given reaction, is
- $$C(s) + H_2(g) + O_2(g) \rightarrow C_{12}H_{22}O_{11}(s)$$
- A) 138.5 B) 155.5 C) 172.5 D) 199.5
38. The percentage by mole of NO_2 in a mixture $NO_2(g)$ and $NO(g)$ having average molecular mass 34 is :
- A) 25% B) 20% C) 40% D) 75%
39. The minimum mass of mixture of A_2 and B_4 required to produce at least 1kg of each product is :
- (Given At. mass of 'A'=10 ; At mass of 'B '=120)
- $$5 A_2 + 2 B_4 \rightarrow 2 AB_2 + 4 A_2 B$$
- A) 2120 gm B) 1060gm C) 560gm D) 1660gm
40. A 5.5 gm sample of an organic compound gave on quantitative analysis 1.4gm of 'N' and 3.6gm of 'C' and 0.5gm of 'H'. Then the empirical formula of the compound is
- A) $C_6H_{10}N_2$ B) C_2H_3N C) C_3H_5N D) $C_4H_6N_2$
41. 5 moles of AB_2 weigh 125×10^{-3} kg and 10 moles of A_2B_2 weight 300×10^{-3} kg.
- The molar mass of $A(M_A)$ and molar mass of $B(M_B)$ in $kgmol^{-1}$ are :
- A) $M_A = 50 \times 10^{-3}$ and $M_B = 25 \times 10^{-3}$
- B) $M_A = 25 \times 10^{-3}$ and $M_B = 50 \times 10^{-3}$
- C) $M_A = 5 \times 10^{-3}$ and $M_B = 10 \times 10^{-3}$
- D) $M_A = 10 \times 10^{-3}$ and $M_B = 5 \times 10^{-3}$

SECTION – II
(MULTIPLE CORRECT ANSWER TYPE)

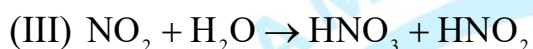
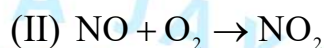
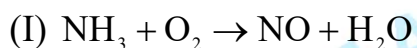
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Marking scheme: +4 for all correct options & +1 partial marks, 0 if not attempted and -2 in all wrong cases

42. The pair of species not having same percentage (mass) of carbon is :



43. Given the following series of reactions :



Select the correct option (s):

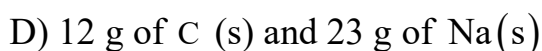
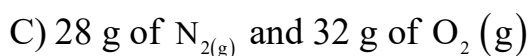
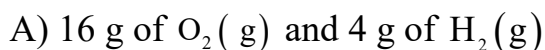
A) Moles of HNO_3 obtained is half of moles of Ammonia used if HNO_2 is not used to produce HNO_3 by reaction (IV)

B) $\frac{100}{6}\%$ more HNO_3 will be produced if HNO_2 is used to produce HNO_3 by reaction (IV) than if HNO_2 is not used to produce HNO_3 by reaction (IV)

C) If HNO_2 is used to produce HNO_3 then $\frac{1}{4}$ th of total HNO_3 is produced by reaction (IV)

D) Moles of NO produced in reaction (IV) is 50% of moles of total HNO_3 produced.

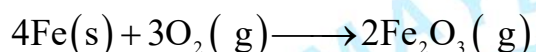
44. Which of the following pairs have the same number of atoms?



45. Which of the following statements about a compound is/are correct ?

- A) A molecule of a compound has atoms of different elements.
- B) A compound cannot be separated into its constituent elements by physical methods of separation.
- C) A compound retains the physical properties of its constituent elements.
- D) The ratio of atoms of different elements in a compound is fixed.

46. Which of the following statements is/are incorrect about the reaction given below:



- A) Total mass of iron and oxygen in reactants = total mass of iron and oxygen in product; therefore it follows law of conservation of mass.
- B) Total mass of reactants = total mass of product; therefore, law of multiple proportions is followed.
- C) Amount of Fe_2O_3 can be increased by taking any one of the reactants (iron or oxygen) in excess.
- D) Amount of Fe_2O_3 produced will decrease if the amount of any one of the reactants (iron or oxygen) is taken in excess.

47. Which of the following statements indicates that law of multiple proportion is being followed.

- A) Sample of carbon dioxide taken from any source will always have carbon and oxygen in the ratio 1:2.
- B) Carbon forms two oxides namely CO_2 and CO , where masses of oxygen which combine with fixed mass of carbon are in the simple ratio 2:1.
- C) When magnesium burns in oxygen, the amount of magnesium taken for the reaction is equal to the amount of magnesium in magnesium oxide formed.
- D) At constant temperature and pressure 200 mL of hydrogen will combine with 100 mL oxygen to produce 200 mL of water vapour.

48. In the formation reaction of NH_3 from N_2 and H_2 , 140 g of N_2 and 40g H_2 were mixed. Select the option(s) which is/are not correct.
- A) Maximum mass of NH_3 which can be formed is 180 g.
B) If % yield of reaction is 80% , then H_2 consumed will be 32 g.
C) Some $\text{N}_2(g)$ will be left after the reaction.
D) If NH_3 formed is 85 g then % yield will be 50%.
49. The sample(s) containing same number of 'Na' atoms as there are 'Na' atoms in 5.3 g of Na_2CO_3 , is/are
- A) 4gm of NaOH B) 6.85gm of NaCl
C) 0.25 mole of Na_2SO_4 D) 5.6gm of Na_3PO_4 .

SECTION – III
(INTEGER ANSWER TYPE)

This section contains 5 questions. The answer is a single digit integer ranging from 0 to 9 (both inclusive).

Marking scheme +3 for correct answer, 0 if not attempted and 0 in all other cases.

50. For the reaction
- $$2x + 3y + 4z \rightarrow 5w$$
- Initially if 1 mole of x, 3 mole of y and 4 mole of z is taken. If 1.25 mole of w is obtained then % yield of this reaction is 'P'. Then $\frac{P}{10}$ is _____
51. Ratio of masses of H_2SO_4 and $\text{Al}_2(\text{SO}_4)_3$ in grams each containing 32 grams of Sulphur is _____
52. The number of significant figures in 50000.020×10^{-3} is _____
53. 4g equimolar mixture of NaOH and Na_2CO_3 contains xg of NaOH and yg of Na_2CO_3 . The value of x is g. (Nearest integer)
54. Iodobenzene($\text{C}_6\text{H}_5\text{I}$) is prepared from aniline($\text{C}_6\text{H}_5\text{NH}_2$) in a two-step process as shown here:
- $$\text{C}_6\text{H}_5\text{NH}_2 + \text{HNO}_2 + \text{HCl} \rightarrow \text{C}_6\text{H}_5\text{N}_2^+\text{Cl}^- + 2\text{H}_2\text{O}$$
- $$\text{C}_6\text{H}_5\text{N}_2^+\text{Cl}^- + \text{KI} \rightarrow \text{C}_6\text{H}_5\text{I} + \text{N}_2 + \text{KCl}$$
- In an actual preparation, 9.30 g of aniline was converted to 16.32 g of iodobenzene ($\text{C}_6\text{H}_5\text{I}$). The percentage yield of iodobenzene is ($I=127$) 'X': Then $\frac{X}{10}$ is _____